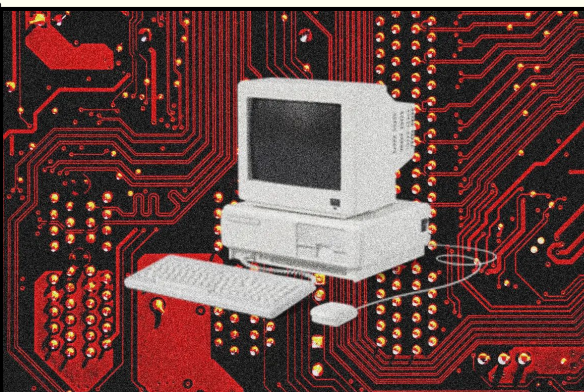
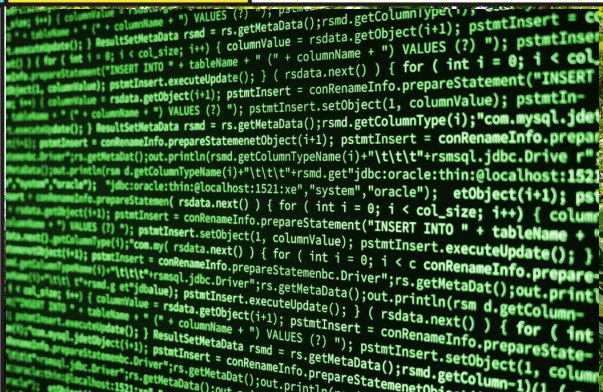


August 5th, 2026

Intro to Java Programing

Agenda

2



What is Object Oriented Programming?

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Java Basics

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Introduce Programing Challenge 1

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Object Oriented Programming

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Code that is organized around “objects” or a digital item that holds information and has actions it can perform.

An example:

Object/Class: Student

Properties: Name, Grade, Student ID, GPA

Actions: Display information about the student, calculate Deans list

You can have multiple **Student** objects that hold unique information:

Student 1: Mike Wazowski, Junior, 1234, 3.7

Student 2: Sully, Junior, 1245, 3.8

Languages that are good for Object Oriented Programming:

- Java
- Python
- C/C++/C#
- Java Script

Java Basics: Data Types

4

Numerical Data Types:

Integer (int): whole numbers, can be (+/-)

1, 3, 4, 10, 100, -25

Double/float (double): decimal or partial numbers

0.0, 2.5, -7.899, 100.001

Long: a type of int that hold up to 21 digits

Short: a type of int that holds at most 5 digits

Written Data Types:

Character (char): a single unicode character within " single quotes

String (str): a "string" of characters combined together and included in "" double quotes.

Can define numbers, letters, symbols or a mix as either a char or string

Other Data Types:

Boolean (bool): holds a *true* or *false* value, good for checking conditions

Conditionals

Conditionals in code follow an If -> Then line of thinking. In Java, we use If/Else Syntax. Conditionals are a way to check for multiple scenarios of the same variable. An example with finding the Letter Grade in a class:

```
int percentGrade = 86;  
string letterGrade = " ";  
  
if(percentGrade >= 90){  
    letterGrade = "A";  
}else if(percentGrade >= 80){  
    letterGrade = "B";  
}else if(percentGrade >= 70){  
    letterGrade = "C";  
}else if(percentGrade >= 60){  
    letterGrade = "D";  
}else{  
    letterGrade = "F";  
}  
System.out.println(letterGrade);
```

Comparison Operators



X == Y

Checks to see if X **EQUALS** Y

(X < Y) || (X > Y)

Checks to see if X is **less than** Y **OR** if X is **greater than** Y

X <= Y && X >= Y

Checks to see if X is **less than or equal to** Y **AND** if X is **greater than or equal to** Y

- == for if two variables (numerical and boolean only) are equal
- > for if one variable is greater than the other
- < for if one variable is less than the other
- >= for if one variable is greater than or equal to the other
- <= for if one variable is less than or equal to the other
- || means OR, when checking at least one condition is met within in one statement
- && means AND, when checking that two statements are met within one statement
- = sets a defined variable to a value

Classes and Objects



```
Class Student{  
  
    private char grade;  
    private String name;  
    private int id;  
    private double gpa;  
  
    #constructor definition:  
    public Student(char studentGrade, String studentName, int  
studentId, double studentGpa) {  
        this.grade = studentGrade;  
        this.name = studentName;  
        this.id = studentId;  
        this.gpa = studentGpa;  
    }  
}
```

#new student object defined as Student1:
Student1 = New Student('J', "Georgia", 1234, 3.75)

—> Class Definition: defines attributes and methods of an object

—> Instance variables: variables present in all **Student** objects

—> Constructor: Initializes a **Student** object with the provided attributes



Programming Challenge 1: Number Guessing Game

**Fill in the starter code
to create the game**

Download the starter code from [github/slack](#) to your computer. Do not edit the Runner class, but you can use it for context clues. Please edit the NumberGuesserStudent.Java file, reading carefully, and filling in all required fields. (hint: use the comments). When you think you have completed the exercise please let a mentor know. If you have questions ask a peer or mentor. NO AI!!!! I will know

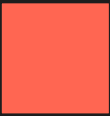


Game Demo



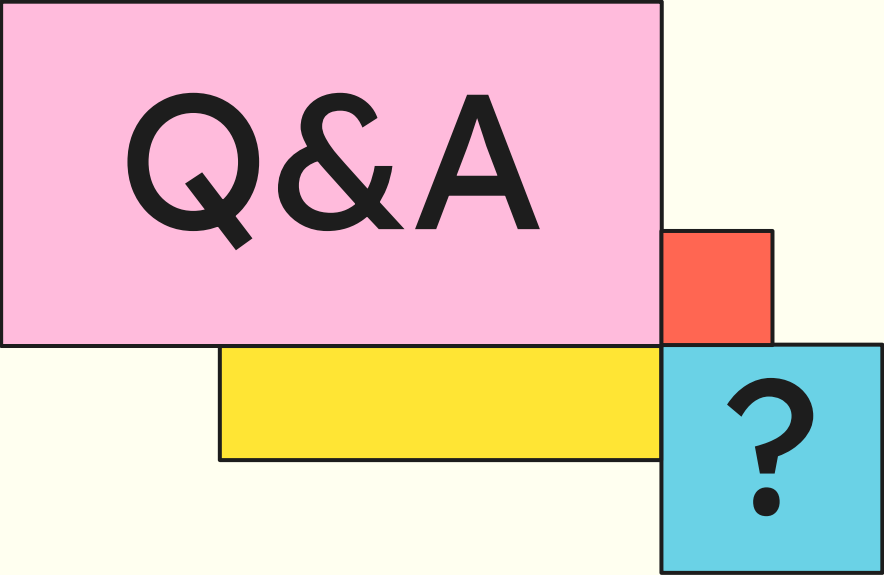


Run Instructions



1. Open a Terminal or PowerShell window
2. Navigate to the directory where your NumberGuesserRun.java and NumberGuesserStudent.java files are located
3. Type: `<javac NumberGuesserRun.java>`, hit enter
4. Type: `<javac NumberGuesserStudent.java>`, hit enter
5. Type: `<java NumberGuesserRun>`, hit enter
6. Play game!

(do not include the `<>` symbols when typing in the terminal)



Q&A

?